

Dependence of economic impacts of climate change on anthropogenically directed pathways

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There are great uncertainties in the projected economic impacts of climate change¹, arising from uncertainties in the climate response², the climate change mitigation pathway³ and the socioeconomic development pathway⁴. Although the relative contributions of these factors are important for climate change related decision-making, they are poorly understood. Here, we show to what extent the projected economic impacts of climate change can be attributed to these three factors. Our modelling framework consisting of global, multisectoral impact models coupled with an integrated assessment model enables us to estimate the global total economic impacts of climate change while incorporating these uncertainty sources. Whereas the most pessimistic pathway without mitigation would result in a net economic impact equivalent to 6.6% (3.9–8.6%) of global gross domestic product at the end of this century, the pathways with stringent mitigation would limit the impact to around or less than 1%. Although the uncertainties are great, the climate change mitigation pathway is the dominant factor and socioeconomic developments can also contribute to alleviate the impacts of climate change. These results suggest that decisions on mitigation and development have a great influence in determining the economic impacts of climate change, regardless of the uncertainties in the climate response.

Information on the total economic impact that will occur with climate change is necessary to inform climate change mitigation and adaptation strategies. However, there are uncertainties in the projected impacts¹ that could hinder decision-making. When considering future climate change impacts, there are several well-recognized sources of uncertainty: the uncertainty in the climate response to increases in greenhouse gas concentrations², the uncertainty in the climate change mitigation (CCM) pathway³ and the uncertainty in the socioeconomic development (SED) pathway⁴. While they are all uncertainties, they arise for different reasons⁵. The first type of uncertainty originates from scientific/technical shortcomings in understanding and simulating the climate system. In other words, it is uncertainty in our knowledge of geophysical processes. On the

other hand, uncertainties in the CCM and SED pathways are associated with how mankind behaves—anthropogenic uncertainties. Although we may not be able to control these pathways precisely⁶, these two factors can in principle be driven by human behaviour. Since they are fundamentally different, the identification and attribution of these sources of uncertainties could inform the decision making of policy makers and motivate choices made by individuals, which together determine the directions of the SED and CCM pathways. The classification of uncertainties used here is different from that used in other areas (for example, aleatoric versus epistemic uncertainty) and does not include all the existing definitions of uncertainties⁷. The purpose of this study is to identify to what extent mankind's behaviour can determine the projected economic impacts of climate change by decomposing sources of uncertainties⁸.

Using various methods, several studies have attempted to estimate the economic impact of climate change^{9–18} but quantitative information on the structure of the uncertainty and on the relative contribution of the above-mentioned factors is limited. Here we quantify to what extent the future economic impacts associated with climate change are determined by anthropogenic factors, represented by differences in the SED and the CCM pathways. To this end, we used a state-of-the-art modelling framework for nine individual sector-specific, climate change impacts (changes in agricultural productivity, undernourishment, heat-related excess mortality, cooling/heating demand, occupational-health costs, hydroelectric generation capacity, thermal power generation capacity, fluvial flooding and coastal inundation due to climate change). Estimated impacts are either directly monetized or fed into a global economic model¹⁹ that translates physical impacts into monetary units (the monetized value as a percentage of gross domestic product (GDP)). The future scenarios are characterized by the combination of five SED pathways, four CCM pathways and five climate models. The SED pathways are represented by the shared socioeconomic pathways²⁰ (SSPs): SSP1, sustainable development; SSP2, middle-of-the-road; SSP3, regional rivalry; SSP4, inequality; and SSP5, fossil fuelled development. Here, SED pathways include not only simple GDP and population changes but also vulnerabilities and exposure (for

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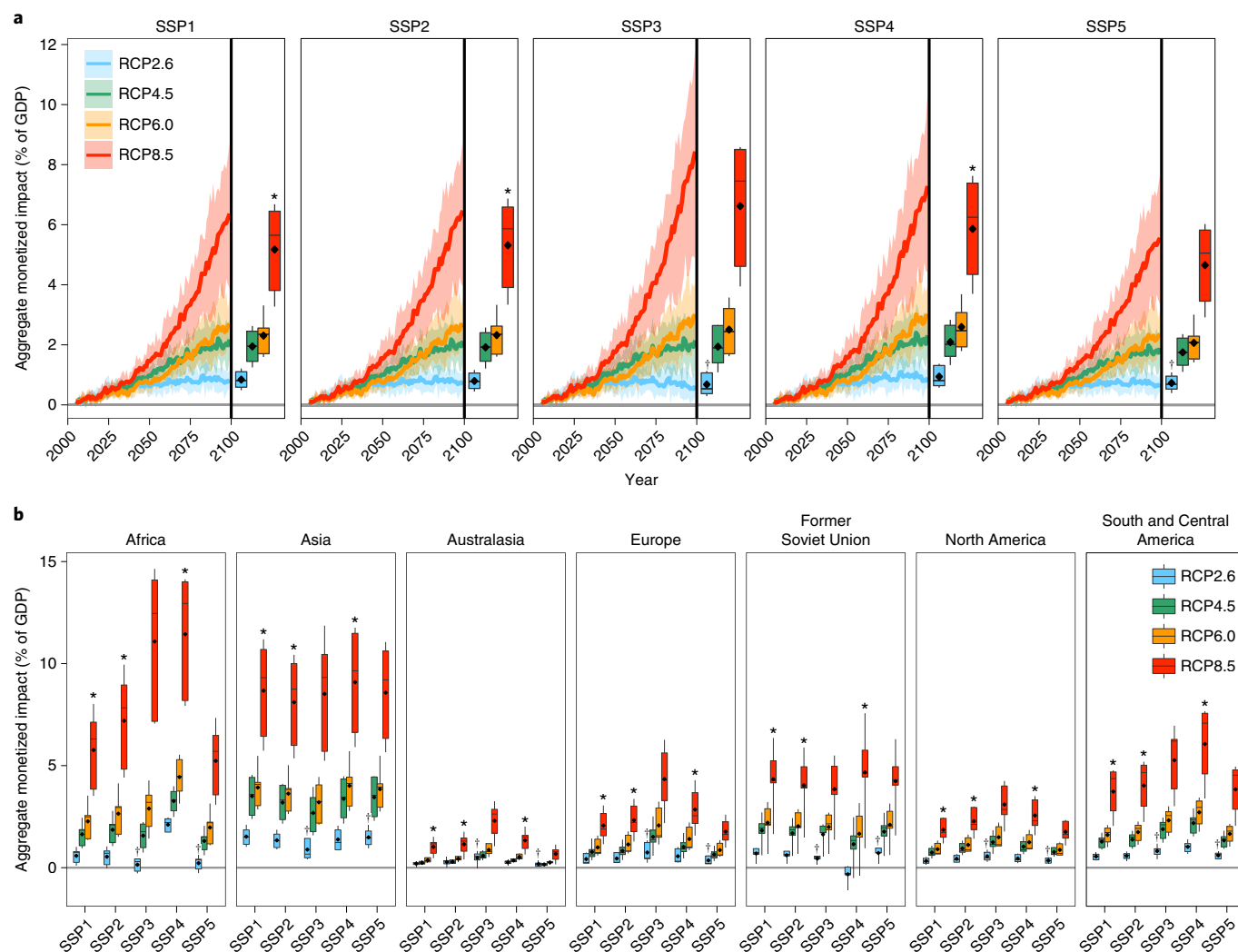


Fig. 1 | Projected economic impact. **a**, Time series of the aggregate monetized impacts and the mean impact for 2080–2099. **b**, Regional impacts. In **a**, the mean (line) and uncertainty range of the five GCMs (shaded area) are shown. For the boxplots in **a** and **b**, the mean impacts for 2080–2099 are shown and the range is the uncertainty range for the five GCMs. Centre line, median; dot, mean; box limits, upper and lower quartiles; whiskers, range. †Achieving the corresponding level of mitigation is less plausible under that SSP. *Reaching the corresponding level of emissions is less plausible under that SSP.

example, economic structures and the number of people living near coastal areas). The CCM pathways are represented by the representative concentration pathways (RCPs) approximately equivalent to global mean temperature rises of 1.7 to 4.5 °C (ref. ²¹; Supplementary Figs. 1 and 2). The results of five different climate models are considered to represent the uncertainty in the climate response. Our extensive modelling and scenario framework allows us to estimate the overall economic impact of climate change in a uniform manner and decompose it into the three uncertainty sources.

There are great differences in the projected economic impacts, depending on the pathway considered (Fig. 1a). Among the plausible SED–CCM combinations, the fragmented SED pathway without CCM policies (SSP3–RCP8.5) results in an economic impact equivalent to 6.6% (3.9–8.6%) of the global total GDP at the end of this century. In contrast, the SED pathway representing the sustainable or middle-of-the-road development scenario with the most stringent CCM pathway (SSP1–RCP2.6 or SSP2–RCP2.6) results in around an eightfold smaller global economic impact—0.8% (0.5–1.2%) for SSP1 and 0.8% (0.4–1.2%) for SSP2. The magnitudes of the impacts vary by region (Fig. 1b) and, in particular, there are great impacts for developing regions such as Asia and Africa under high-emission

scenarios that can be modified by the SED pathways. In general, all regions are more adversely affected under the fragmented-world scenario (SSP3) and the adverse effects are concentrated in developing regions under the inequality scenario (SSP4).

Most of the uncertainty (variance) in the projected impacts is attributable to anthropogenic factors (SED and CCM pathways), particularly in the latter half of the twenty-first century, and the contribution of the CCM pathway is the greatest (Fig. 2). The uncertainty in the first quarter of the study period is dominated by the climate sensitivity, which suggests that the effects of changes in the SED and CCM are less critical and discernible over shorter time scales. Contrastingly, the contribution of the anthropogenic factors (particularly CCM pathways) to the uncertainties becomes important over time because of the gradually increasing divergence in the pathways and their cumulative effects. These tendencies are also observed on the regional scale but the relative contribution of the SED pathways is not negligible in some regions (for example, Africa, Australasia and Europe).

We also identified the relative contributions of each uncertainty source for each sector (Fig. 3). The effects on the precipitation-dependent impacts (for example, fluvial flooding and hydroelectric

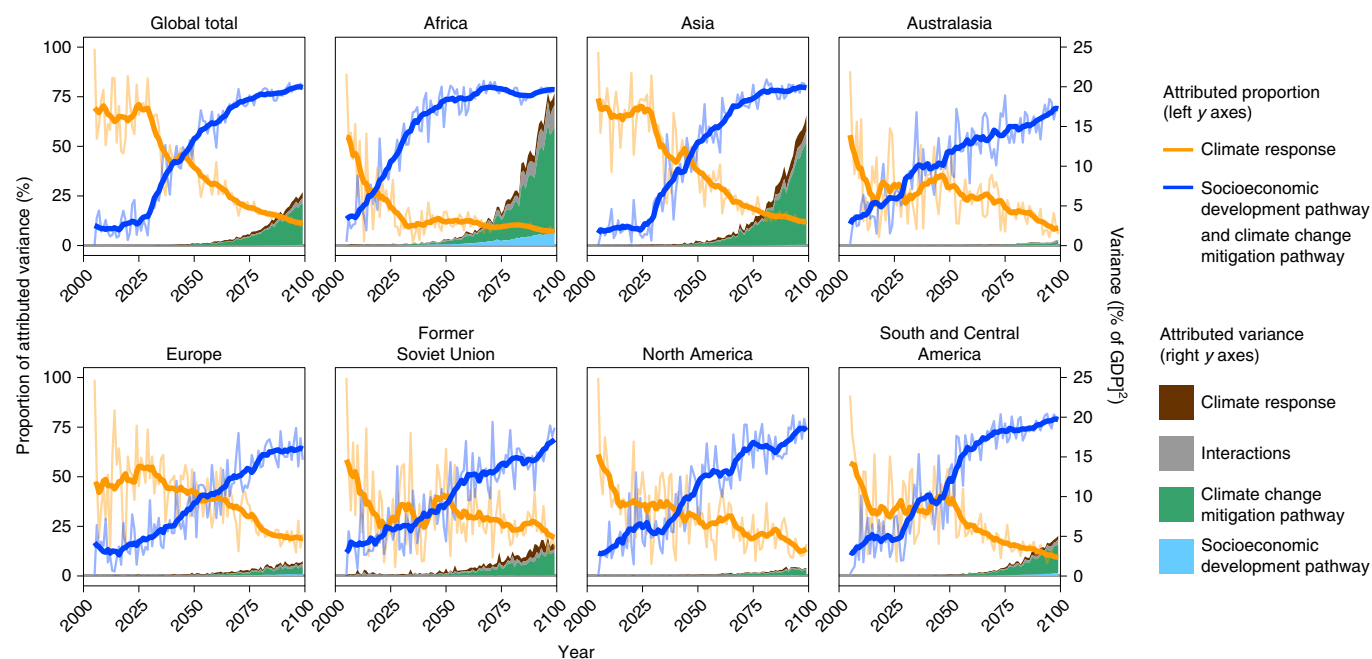


Fig. 2 | Variance of the projected impact attributed to each factor. The stacked plots (referring to the right y axes) are the attributed variance and the lines (referring to the left y axes) are the proportion of the attributed variance to the total variance. The bold lines are the 10-yr moving averages and the thinner lines are the raw values. The variances attributable to differences in GCMs, RCPs and SSPs by the ANOVA are referred to as the climate sensitivity, CCM pathway and SED pathway, respectively.

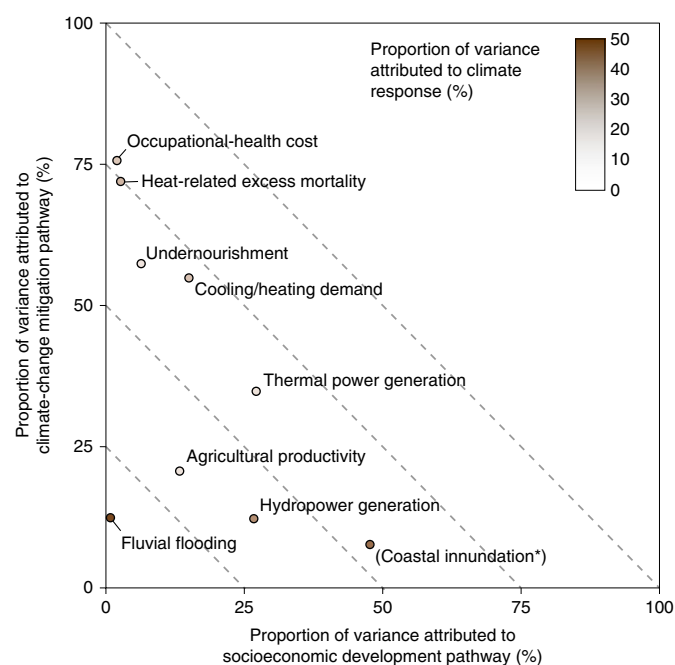


Fig. 3 | Proportion of attributed variance for sectoral impacts. Variances attributed to differences in GCMs, RCPs and SSPs by the ANOVA are referred to as the climate sensitivity, CCM pathway and SED pathway, respectively. The positions of the dots represent the proportion of the variance attributed to SED and CCM pathways. Colours represent the proportion of variance attributed to the climate response. These values are based on the results for 2080–2099. *The result for the coastal inundation does not include the results from the five GCMs due to lack of data.

power generation) are sensitive to the uncertainty in the climate response. This is because it is more difficult to predict regional-scale changes in the precipitation with the general circulation models (GCMs) than temperature². However, in terms of the economic impact, the impact of these sectors is less than that of the other sectors (Supplementary Figs. 3 and 4). Thus, the aggregated economic impacts are less sensitive to uncertainty in the climate response.

These projected economic impacts are within the range of those found in existing studies²³ but, due to our coverage of a wide range of scenarios, our results provide useful insights, particularly about the changes in vulnerability arising from differences in the SED pathways. While the contribution of the CCM pathway is the dominant factor in determining the impact of climate change, the SED pathway can amplify or dampen the impact even under the same climatic conditions by altering the temperature–impact relationship, particularly when the temperature rise is large (Fig. 4a). Under the lower development scenarios, the impacts are larger. The differences in the regional economic vulnerability to temperature increase can be explained by two factors (Fig. 4b). First, the baseline temperature is related to the magnitude of the impacts: if the baseline temperature is already high, the same temperature increase yields greater impacts than at a lower baseline temperature. This is quantitatively consistent with the econometrically estimated spatial pattern of impacts reported elsewhere¹⁶, although it is less robust in sensitivity analyses. Second, SED (increase in per capita GDP) alleviates the adverse effects of climate change by reducing vulnerability. The concavity of the fitted surface suggests that economic development in warmer developing countries can reduce vulnerability more effectively but additional development in developed countries is not as effective. Nevertheless, it should also be noted that the SED alone does not mitigate the effects of climate change without the stringent CCM.

In this study, we demonstrated that most of the uncertainty in the projected economic impacts was from anthropogenic sources

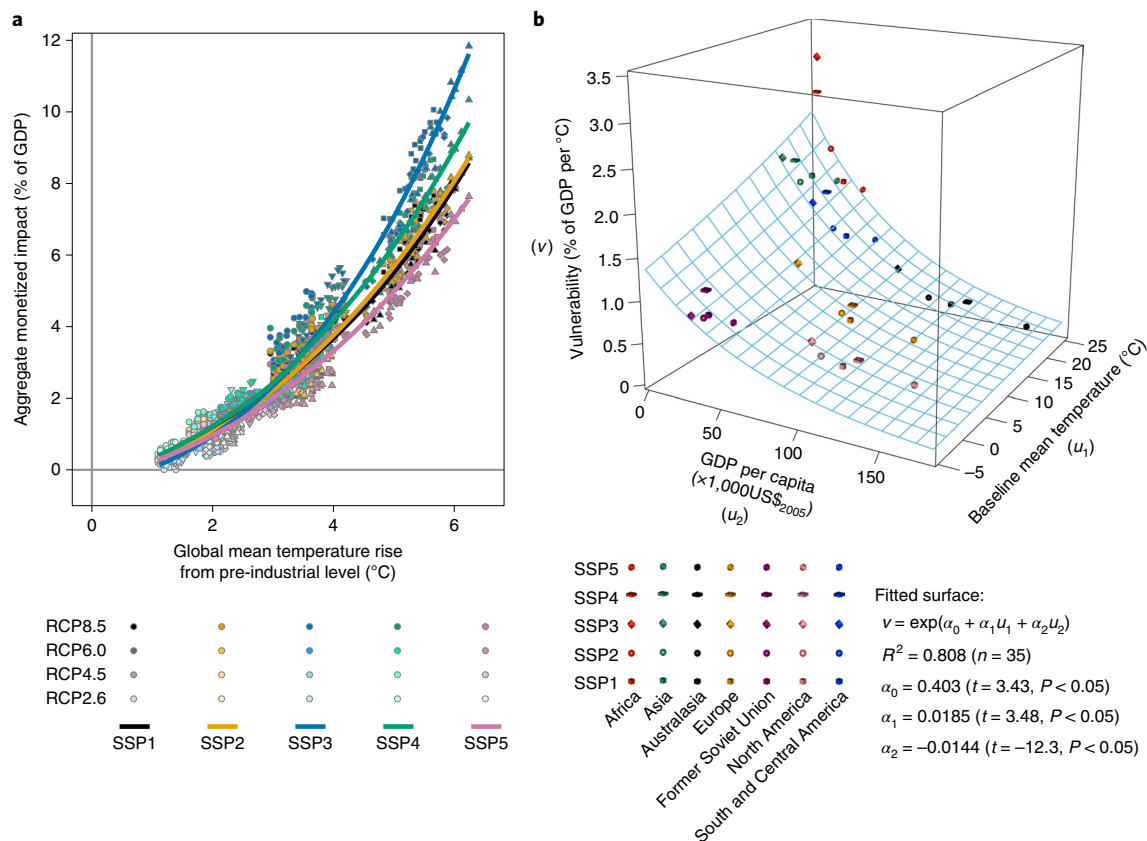


Fig. 4 | Relationships of impact, vulnerability, temperature and SED. **a**, Relationship between global mean temperature increase and the projected impact. **b**, Relationships of regional vulnerability, baseline temperature and GDP per capita. In **a**, annual results for 2080–2099 are plotted. Curves are fitted by local polynomial regression. Shapes of the points represent GCMs. In **b**, the vulnerabilities are defined as the slopes of the regression lines shown in Supplementary Fig. 5. The vulnerability and GDP per capita are based on the values for 2080–2099. The baseline temperature is based on the values for 2001–2010. The statistical test is two-tailed.

and that the SED could alleviate vulnerability to increased temperature. Several factors may distort these conclusions. First, the contributions of different sectors are not equal (Supplementary Figs. 3 and 4). If a certain sector dominates the overall impact, and the projections for this dominant sector are not sufficiently accurate, the overall conclusion can be equivocal. We investigated this possibility and confirmed that our conclusions were not due to a single dominant sector (Supplementary Discussion 1). Second, we used all possible combinations of SSPs and RCPs, some of which are thought to be implausible. Even if we exclude implausible combinations, the same conclusions are obtained (Supplementary Discussion 2). Third, we used a limited number of GCMs and do not cover the full range of uncertainty in the climate response²² but, again, our conclusions would hold even if the full range of uncertainty were included (Supplementary Discussion 3). Fourth, there is another well-recognized uncertainty source: the uncertainty associated with the variation between the impact models^{24,25}. The different models predict different levels of impacts and thus there is the possibility of over/underestimation of impacts. However, incorporating multiple impact models into our framework is beyond the scope of this study. Therefore, instead we investigated the effect of over/underestimation for each impact model and found that a moderate amount of over/underestimation of each sector's impact did not affect the overall conclusions (Supplementary Discussion 4). While we used only one impact model for each sector, further model intercomparison using a similar approach to this study will provide meaningful insights. In summary, our main conclusions that most of the uncertainty in the projected economic impacts was from anthropogenic

sources and that the SED could alleviate vulnerability to increased temperature are robust even under the above concerns.

Our modelling framework is essentially based on the extension or extrapolation of current social conditions and does not consider fully drastic structural innovations in society. This is one of the most important unmet challenges when conducting long-term future projections²⁶. However, this also means that there is further space to be explored both in the CCM and SED pathways and the contributions of these societal changes could be underestimated. This would strengthen, rather than contradict, our main conclusion that mankind's decisions relating to societal changes have a great influence on determining the economic impacts of climate change.

We used the percentage of GDP as an impact indicator but we do not claim it is the best or only indicator to evaluate the impacts of climate change. The results of this study do not indicate that less-affected regions' mitigation efforts are of less importance. Since the cost and possible side effects of CCM are not included, the results of this study alone do not justify the stringent mitigation pathways corresponding to the 2 or 1.5°C goals. We considered all possible combinations of discrete SSP and RCP scenarios, which are thought to cover the full range of future pathways, although there are infinite possible pathways²⁷ (for example, an intermediate pathway between SSP1 and SSP2). Identifying the optimal pathway is beyond the scope of this study and more studies are needed, including discussion on the definition of justice and how optimal is defined.

Nevertheless, the results indicate that the projected economic impacts of climate change are mainly determined by human decisions and actions and the CCM pathway is the dominant factor.

Regardless of the SED pathways, the impacts of climate change cannot be fully mitigated without adopting the stringent CCM. However, the SED cannot be neglected, particularly when we will fail to reduce greenhouse gas emissions sufficiently. Moreover, successful adoption of the stringent CCM strategy highly depends on the SED pathway and thus our choice of SED pathway, which includes a wide range of social changes^{20,28–30}, can be regarded as a necessary measure to achieve CCM. In this context, climate change should be interpreted as a broad sustainable development issue that is ultimately dependent on the consequences of the aggregation of the actions and decisions of billions of individuals and millions of organizations.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of code and data availability and associated accession codes are available at <https://doi.org/10.1038/s41558-019-0578-6>.

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Author contributions

S.F., N.H., T.H., Y.Hirabayashi, Y.Honda, T.I., N.K., C.P., Z.S., K.Takahashi, J.T., M.Tamura, M.Tanoue, K.Tsuchida, H.Y. and Q.Z. conducted analyses on the sectoral impacts and provided the data. J.T. conducted the analysis of the aggregated impacts. J.T. and S.F. wrote the manuscript. T.O. and Y.Hijioka directed the study. All authors participated in the interpretation of the results, discussion and revising the draft of the manuscript.

Competing interests

J.T. was employed by Toshiba Corporation, which is associated with the manufacture, sale, distribution and marketing of hydro/thermal power plants, until February 2016. K.Tsuchida has been employed by Nippon Koei, which is associated with consultation on natural disaster prevention (including fluvial flooding and coastal inundation) and on hydro/thermal power plants, since April 2019. The other authors declare no competing interests.

Additional information

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Methods

Overall framework. To obtain projections of the overall impacts of climate change, we adopted a bottom-up approach by conducting process-based impact simulations for each modelled sector and aggregating the monetized impacts. We incorporated as many sectors as possible whose impacts could be monetized. This included the impact of changes in (1) agricultural productivity, (2) undernourishment, (3) heat-related excess mortality, (4) cooling/heating demand, (5) occupational-health cost, (6) hydroelectric power generation capacity, (7) thermal power generation capacity, (8) fluvial flooding and (9) coastal inundation, due to climate change. The impacts were originally calculated for a $0.5^\circ \times 0.5^\circ$ grid for each year during the twenty-first century and the calculated impacts were aggregated to 17 regions unless otherwise noted (above, results are further aggregated into seven regions). Then, these sectoral impacts (except for fluvial flooding, coastal inundation and heat-related excess mortality) were supplied as input to the Asia Pacific integrated model/computable general equilibrium (AIM/CGE) model. The AIM/CGE model was then used to calculate changes in the GDP associated with these impacts. For undernourishment, heat-related excess mortality and fluvial flooding, the non-market values of lives lost represented by willingness-to-pay to avoid these risks were also incorporated. For fluvial flooding and coastal inundation, the economic impacts were directly monetized without using the AIM/CGE model but they include both direct and indirect impacts. Autonomous adaptation to climate change was also incorporated. Here, autonomous adaptation refers to, for example, reduction of vulnerability due to increase in the income level, changes in the industrial structure and the economy's response to the shock through the market mechanisms represented by the AIM/CGE model. Although we covered most of the major sectors that will probably be affected by climate change, there also remain sectors such as conflict³¹ and crime³², and the causal relationship between these sectors and climate change is controversial. The consequences of catastrophes were also not considered. Incorporation of these sectors is the next step in future work.

Scenario sets. We used the SSP/RCP scenario matrix. The SSP represents the SED pathway³ and we used SSP1, SSP2, SSP3, SSP4 and SSP5. The RCP scenarios represent the CCM pathway³ and we used RCP2.6, RCP4.5, RCP6.0 and RCP8.5. While not all of the combinations of SSPs and RCPs are equally plausible³⁰, we considered all SSP $\times$ RCP combinations to enable the decomposition of the effects by factor. To incorporate the uncertainty in climate change projections, climate data from five different GCMs were used, namely, GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, MIROC-ESM-CHEM and NorESM1-M. When simulating the impacts of climate change in each sector, the bias-corrected output of the GCMs^{33,34} was used. Therefore, we have $5 \times 4 \times 5$ scenario runs for each study unless otherwise noted. We also ran simulations under no climate change conditions, in which climate conditions were fixed at present-day levels, for each SSP. The differences in the effects between the results generated under climate change conditions and no climate change conditions were regarded as the impacts of climate change.

Agricultural productivity. Changes in crop yields associated with climate change are computed with the crop yield growth model with assumptions (CYGMA) on climate and socioeconomy^{35,36}, which is a biophysical global gridded crop model that can explicitly consider changes in agronomic technology and management driven by economic growth as well as changes in the biophysical response of a crop to environmental conditions. The yields of major crops (rice, wheat, soybean and maize) were based on the estimates from CYGMA (<https://doi.org/10.20783/DIAS.529>). For other crops (for example, sugar), we used the climate-change impacts on yield estimated by the Lund–Potsdam–Jena managed land dynamic global vegetation and water balance model³⁷. The grid-cell simulated yields were aggregated into regions and input to the AIM/CGE model³⁸.

Undernourishment. Climate change will affect food prices and impoverished people's access to food. We quantified the impacts of climate change on undernourishment and related negative health impacts using two economic metrics³⁹. First, morbidity and mortality, caused by being underweight as a child and the associated economic loss, were computed with the AIM/CGE model with consideration of changes in the labour force, population and demands for healthcare. The prevalence of undernourishment, mortality and morbidity were estimated on the basis of the calculated income distribution and food price. Second, changes in mortality were assessed economically in the form of the value of lives lost. For this purpose, the value of statistical life (VSL), a summary measurement of the willingness-to-pay for mortality risk⁴⁰, was used. A reference value of the VSL (data from China) was applied to other regions and periods considering income elasticity according to the Organisation for Economic Co-operation and Development's guidelines^{39,40}. Then, the sum of the VSL was derived by multiplying the VSL and the mortality rate in each region.

Heat-related excess mortality. The impact of climate change on heat-related excess mortality was estimated using the method developed by Honda et al.⁴¹ and the associated economic effect was evaluated by the values of lives lost represented by the VSL^{39,40} as explained in the section Undernourishment.

The temperature–mortality relationship forms a V-shaped curve. The difference in mortality between the optimal temperature (the temperature at which mortality becomes lowest) and temperatures outside of the optimal temperature is defined as the excess mortality. In the projections, we considered the annual mean mortality for the population over 65-years-old and changes in the daily maximum temperature. Temperature–mortality relationships were determined separately for each grid on the basis of the baseline climate conditions. Additional acclimatization to future warming in the temperature–mortality relationship is not considered. Since only the mortality for people over 65-years-old is considered, we did not calculate the impact on the real economy through changes in the labour force.

Cooling/heating demand. The economic impact of changes in cooling/heating demand^{42,43} considers both the cost of cooling/heating energy use and the cost of heating/cooling device installation and management using the AIM/CGE model. Cooling degree days (CDD) and heating degree days (HDD) were calculated with the output of the GCMs and used as the climatic driver of the cooling/heating demands. Energy demand for cooling/heating was calculated as a function of CDD/HDD, labour force or population, floor area and cooling/heating device penetration rate⁴⁴. These values were directly taken from the SSP scenarios or estimated on the basis of the available data. Adoption of different kinds of cooling/heating technologies and their costs were determined using the AIM/end-use database⁴⁵. We used different assumptions for the autonomous energy efficiency improvement and the threshold for CDD/HDD among SSPs according to their storylines⁴³.

Occupational-health cost. The occupational-health cost⁴⁶ was defined as the economic loss caused by reduction of the per-hour workable time (labour capacity) due to workers' exposure to heat stress. We estimated the wet bulb globe temperatures⁴⁷ for work sites from the output of the GCMs. The labour capacity was calculated from the estimated wet bulb globe temperature and the intensity (metabolic rate) of the work based on occupational-health recommendations⁴⁸ and used as the labour productivity coefficient in the AIM/CGE model. We assumed that the recommendations for labour capacity limitations are strictly followed. Different assumptions for the work location (indoor or outdoor) and the metabolic rate of work were used for the different industrial sectors. For industrial sectors where work is assumed to be conducted indoors, the penetration rate of air conditioners was also considered. To estimate the future penetration rate of air conditioners, we adopted the same method used in the cooling/heating demand study⁴⁴.

Hydroelectric power generation capacity. The hydroelectric power generation capacity will be affected by changes in precipitation and will influence the economy through changes in the cost of electricity generation and the prices for consumers⁴⁹. The grid-based theoretical hydroelectric power potential (THP) was projected using the H08 global hydrological model^{50,51}. THP (J) was defined as $THP = Q \times g \times h$, where Q is the annual streamflow or mass of water (kg), g is the gravitational acceleration (m s^{-2}) and h is the height difference between the cell in question and the adjacent downstream cell into which the river flows (m). Regional variations in THP were assumed to lead to the regional HG potential variation, which is used as the upper bound on the hydropower generation capacity in the AIM/CGE model⁴⁹.

Thermal power generation capacity. Cooling water shortages lead to changes in the usable thermoelectric capacity. Climate change increases the probability of lower river flow, which then increases the probability of a reduction in the usable capacity. We used the H08 global hydrology model^{50,51} to calculate the magnitude of regional usable capacity changes attributable to cooling water shortages⁵². In this study, we measured changes in the generation capacity using the cooling water sufficiency index, which is calculated as the ratio of the yearly total cooling water abstraction to yearly total cooling water demand. Regional variations in the cooling water shortage index were assumed to lead to the variation in the regional thermal power generation potential, which is used as the capital productivity of the thermal power generation sector in the AIM/CGE model⁵³.

Fluvial flooding. The consequences of fluvial flooding (economic loss and mortality) were calculated as the exposure multiplied by the vulnerability⁵⁴. Mortality was monetized by multiplying the VSL by the mortality rate as explained in the undernourishment section. Flood exposures were calculated as the gridded GDP and population exposed to the projected annual maximum inundation extent. The vulnerability was calculated as a function of the income level and time estimated from the historical data⁵⁵. The inundation extent was calculated using a global river and inundation model simulation (CaMa-Flood)⁵⁶ at $0.25^\circ \times 0.25^\circ$ horizontal resolution, driven by daily runoff data obtained from a land surface model (MATSIRO)⁵⁷ simulation at $0.5^\circ \times 0.5^\circ$ resolution. The result was then downsampled to $5' \times 5'$ using the method of Ikeuchi et al.⁵⁸. A gridded map of the population was created by distributing the country-level population from the SSP scenarios using a $5' \times 5'$ population map (HYDE3.1)⁵⁹. A gridded GDP map was calculated from the population multiplied by the GDP per capita for the SSP scenarios, under the assumption that the GDP per capita does not vary within a country.

Coastal inundation. Inundation damage in coastal zones⁶⁰ will be one of the most serious consequences of sea level rise. Inundated areas and temporal changes in inundation were estimated using topographic data (ETOPO1), astronomical high tides (mean higher height water levels) data and sea surface height data from four GCMs (HadGEM2-ES was not included due to data availability), adjusted vertically at the geoid. The ETOPO1, which provides elevation data on land (metres above mean sea level) and water depth at 1 arcmin and is a global relief model of the Earth's surface that integrates land topography and ocean bathymetry, was averaged onto a 2.5 arcmin gridded resolution. A macro-estimation method was used to evaluate the economic impact of inundation. Economic damage estimates were based on the econometric relationship between past hydrological disasters, the affected population and per capita GDP^{61,62} using the Centre for Research on the Epidemiology of Disasters emergency events database⁶³ and the World Bank's national statistics. This estimation only includes the effects of gradual sea level rise and acute events such as storm surges were not considered. We could simulate the impact of the sea level rise only for SSP1, SSP2 and SSP3 due to data availability. To obtain estimates for SSP4 and SSP5, the results were extrapolated by constructing and using emulators (see Supplementary Methods 3).

AIM/CGE model. The AIM/CGE model is a global-scale economic model based on general equilibrium theory¹⁹. The AIM/CGE model has been widely applied to CCM and impact studies. In the model, the world is divided into 17 regions (Supplementary Table 3) and each region has about 40 industrial sectors (Supplementary Table 4) and a household sector. Production for each industrial sector is represented with multinested production functions, which receive labour, capital, land, energy and materials as input. The preference of consumers (household sector) is expressed with a utility function. Industrial sectors maximize their profits and the household sector maximizes its utility under budget constraints. International trade, taxes and capital formation/depreciation are also considered in the model. Calculations are conducted at yearly intervals recursively from the base year (2005) until the end of the twenty-first century. The model parameters are calibrated on the basis of a social accounting matrix for the base year (2005), then updated every year according to the SSP storyline. In this study, we did not place constraints on the greenhouse gas emissions when running the AIM/CGE model and thus the interaction between climate change impact and CCM is not considered. However, the potential effects of the interaction between climate change impacts and mitigation are miniscule compared to the aggregated total economic impact^{49,64}. The interactions between impacts among sectors were also not considered. These interactions will be of interest particularly when considering multiple specific indicators but are beyond the scope of this study. The full model description is available in the model manual¹⁹ and how to incorporate SSP storylines into the model is described in a published paper⁶⁵. The ways to integrate the sectoral impacts and the AIM/CGE model are described in their respective articles^{18,39,42,46,49,53} and brief overviews of the frameworks are provided in Supplementary Methods 2.

Analysis of the aggregated results. The estimated economic impacts were aggregated in US\$ (2005) at market exchange rates for seven regions (Africa, Asia, Europe, Former Soviet Union, North America, Oceania and South and Central America) and for the global total impact. We used the percentage of GDP as the indicator for the degree of the economic impact and we calculated this indicator for all scenarios across the nine sectors. Missing economic impact data due to the unavailability of the corresponding GCM results are imputed by the EM algorithm⁶⁶ to enable the subsequent analysis.

The contributions of uncertainties in the climate response, SED pathway and CCM pathway to the uncertainty in the aggregated economic impacts were quantified using the analysis of variance (ANOVA). Let $X_{\text{ssp,rcp,gcm}}$ represent the economic impact in a certain region and year under SSP (ssp), RCP (rcp) and GCM (gcm). The variance is represented by the sum-of-squares of the deviation from the mean and the total sum of squares (SS_{TOTAL}) is calculated using equation (1).

$$SS_{\text{TOTAL}} = \sum_{\text{ssp}} \sum_{\text{rcp}} \sum_{\text{gcm}} (X_{\text{ssp,rcp,gcm}} - \bar{X})^2 \quad (1)$$

Here, $\bar{X}$ is the pooled mean of $X_{\text{ssp,rcp,gcm}}$. Then, SS_{TOTAL} can be decomposed into SS_{SSP} , SS_{RCP} , SS_{GCM} and SS_{INT} ($SS_{\text{TOTAL}} = SS_{\text{SSP}} + SS_{\text{RCP}} + SS_{\text{GCM}} + SS_{\text{INT}}$).

$$SS_{\text{SSP}} = N_{\text{RCP}} N_{\text{GCM}} \sum_{\text{ssp}} (\bar{X}_{\text{ssp}} - \bar{X})^2 \quad (2)$$

$$SS_{\text{RCP}} = N_{\text{SSP}} N_{\text{GCM}} \sum_{\text{rcp}} (\bar{X}_{\text{rcp}} - \bar{X})^2 \quad (3)$$

$$SS_{\text{GCM}} = N_{\text{SSP}} N_{\text{RCP}} \sum_{\text{gcm}} (\bar{X}_{\text{gcm}} - \bar{X})^2 \quad (4)$$

$$SS_{\text{INT}} = SS_{\text{TOTAL}} - SS_{\text{SSP}} - SS_{\text{RCP}} - SS_{\text{GCM}} \quad (5)$$

N_{SSP} , N_{RCP} and N_{GCM} represent the numbers of SSPs, RCPs and GCMs, respectively. $\bar{X}_{\text{ssp}}$, $\bar{X}_{\text{rcp}}$ and $\bar{X}_{\text{gcm}}$ are the averages over ssp, rcp and gcm, respectively. This means that differences among the total variance (SS_{TOTAL}), SS_{SSP} , SS_{RCP} and SS_{GCM} are caused by the differences in SSPs, RCPs and GCMs. Here, SS_{SSP} and SS_{RCP} can be interpreted as the uncertainties arising from the choice of SED and the CCM pathways. On the other hand, SS_{GCM} arises from the uncertainty in the climate response including both inaccuracies in the representation of the climate system in the models and the chaotic nature of the climate system. SS_{INT} is caused by interactions (combinations) between these factors. Therefore, the relative contributions of these factors are represented by $SS_{\text{SSP}}/SS_{\text{TOTAL}}$, $SS_{\text{RCP}}/SS_{\text{TOTAL}}$, $SS_{\text{GCM}}/SS_{\text{TOTAL}}$ and $SS_{\text{INT}}/SS_{\text{TOTAL}}$, respectively. To avoid possible biases due to the different number of levels of each factor, we further used a subsampling technique⁸ (Supplementary Methods 4).

The regional vulnerability was defined as the expected change in the economic impact (% of regional GDP) caused by a 1 °C regional temperature increase. These values were calculated from the slope terms in the linear regression equations (Supplementary Fig. 5). The calculated vulnerability for each region and SSP ($v_{\text{ssp,region}}$) was regressed using the following equation (6).

$$v_{\text{ssp,region}} = \exp(\alpha_0 + \alpha_1 u_{1,\text{ssp,region}} + \alpha_2 u_{2,\text{ssp,region}}) + \varepsilon_{\text{ssp,region}} \quad (6)$$

Here, $u_{1,\text{ssp,region}}$ and $u_{2,\text{ssp,region}}$ are the mean land-area temperature in the base-period (2001–2010) and the per capita GDP in a region under SSP, respectively ($\varepsilon_{\text{ssp,region}}$ is the error term).

Robustness of the results. To check the robustness of our conclusions, we conducted the following sensitivity analysis.

First, we checked whether the overall result was dominated by a single sector. For this purpose, we conducted a leave-one-sector out aggregation. When aggregating the economic impact, the result for one sector was excluded and the results of the remaining eight sectors were used. This procedure is repeated nine times changing the excluded sector. For each repetition, the same analyses (ANOVA and vulnerability regression) were conducted and we checked whether the main conclusions would hold if any one sector's result were excluded.

Second, we checked whether our conclusions arose from implausible combinations of SSP and RCP. Not all combinations of SSP and RCP scenarios are equally plausible. Thus, we excluded unlikely combinations of SSP and RCP and conducted the same analyses (ANOVA and vulnerability regression).

Third, we investigated the sensitivity of the results to the choice of GCMs used in this study. The five GCMs used in this study³³ do not cover the full range of uncertainty in the climate response. The five GCMs cover approximately 50–90% of the range of uncertainty in the temperature and 30–80% in the precipitation, depending on the region considered²². To calculate their effect on the variance of the estimated impacts, each regional impact was regressed on the changes in temperature and precipitation. Then, on the basis of the coefficients in the regression formulas, the expected inflation of variance was calculated (see Supplementary Discussion 3).

Fourth, we investigated the effect of over/underestimation of each sector's impact. To a certain degree, over/underestimation of the impact is unavoidable. We cannot identify which sectors' impacts are underestimated and which sectors' impacts are overestimated. Therefore, there are many possible combinations of over/underestimation for the sectoral results. We adopted a Monte Carlo approach to cover the possible combinations. The total impact is expressed as follows:

$$X_{\text{ssp,rcp,gcm}} = \sum_{\text{sector}} X_{\text{sector,ssp,rcp,gcm}} \quad (7)$$

Here, $X_{\text{sector,ssp,rcp,gcm}}$ is the economic impact of the corresponding sector. If we consider over/underestimation of each sectoral result, it can be rewritten as:

$$X'_{\text{ssp,rcp,gcm}} = \sum_{\text{sector}} \beta_{\text{sector}} X_{\text{sector,ssp,rcp,gcm}} \quad (8)$$

where β_{sector} is the coefficient of over/underestimation. We simulated $X'_{\text{ssp,rcp,gcm}}$ 10,000 times by changing the values of β_{sector} at random (using a logarithmically uniform distribution) within the range [1/4, 4]. The factor 4 was determined from the results of a recent meta-analysis²³. For each simulation, we conducted the same analyses (ANOVA and vulnerability regression).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Data required to reproduce the main results are available in the Supplementary Datasets and Code.

Code availability

Computer code required to reproduce the main results are available in Supplementary Datasets and Code.

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State explicitly what error bars represent (e.g. SD, SE, CI)

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Data collection

No software was used to collect data.

Data analysis

R (Version 3.4.3, Windows, 64bit) was used with custom codes.

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Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study was based on the output of models. Three factors (socioeconomic conditions, climatic conditions, and climate models) were considered.
Research sample	Data was obtained from the results of existing studies (reference No. 38,39,41,42,43,46,49,53,54,60) and their extended studies.
Sampling strategy	This study did not involve statistical sampling procedure.
Data collection	Data was provided by the co-authors who conducted individual model analysis.
Timing and spatial scale	Data (model output) was obtained between the period of 2005-2099 with 1-year temporal resolution. Spatially, the world is divided into 7 regions.
Data exclusions	No data were excluded from the analysis.
Reproducibility	Data and computer code to reproduce the results are attached as supplementary data. We confirmed they could reproduce the results.
Randomization	Allocations were not conducted in this study.
Blinding	This study is a model-based study and blinding was not applicable.
Did the study involve field work?	☐ Yes ☒ No

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☒	☐ Unique biological materials
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging